Madhusudhan Reddy Pittu

Curriculum Vitae

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mathrulestheworld.github.io

Summary

- Research focus: **AI safety and alignment**, in particular why narrow fine-tuning changes model behavior broadly, and how to predict and prevent it. Selected for the **Iliad Fellowship** in AI alignment (2026).
- Postdoctoral Fellow at Georgia Tech ISyE with Mohit Singh; previously a Postdoctoral Associate at NYU's Courant Institute with Anupam Gupta.
- Ph.D. in Computer Science from CMU; publications at STOC, FOCS, SODA, and ICALP, including optimization from pairwise comparisons, the feedback model behind RLHF.
- Bronze Medalist representing India at the 57th International Mathematical Olympiad (IMO).

AI Safety Research

Summer 2026–present **LLM Fine-Tuning, Generalization, and Alignment**, *Ongoing, unpublished*

Self-directed study of how narrow fine-tuning changes behavior far outside the training task.

- Reproduced emergent misalignment and weird generalization, and built new model organisms.
- Built evaluations with held-out questions, matched controls, and blinded LLM judges.
- Developed a gradient-based predictor of the shift and regularizers that partly reduce it.
- Now testing whether preference learning (DPO, PPO/GRPO with reward models) causes the same spillover.
- Working with a group from Google Research, NYU, and Northwestern on a geometric theory of fine-tuning spillover.

Skills

Theory — Algorithm design and analysis, combinatorial optimization, linear and integer programming, convex optimization, linear algebra, probability and statistics.

AI safety — Model organisms, behavioral evaluations with blinded LLM judges, LLM fine-tuning and post-training (LoRA, DPO, PPO/GRPO).

Tools — Python, PyTorch, Hugging Face Transformers, vLLM, Slurm GPU clusters, C++.

Education and Experience

Sept.–Dec. 2026 **Postdoctoral Fellow**, *Georgia Institute of Technology*, Atlanta
H. Milton Stewart School of Industrial and Systems Engineering (ISyE)
Host: Prof. Mohit Singh

2025–2026 **Postdoctoral Associate**, *Courant Institute, New York University*, New York
Host: Prof. Anupam Gupta

2021–2025 **Ph.D. in Computer Science**, *Carnegie Mellon University*, Pittsburgh
Advisors: Prof. Anupam Gupta and Prof. David P. Woodruff

2017–2021 **B.Tech. in Computer Science and Engineering**, *Indian Institute of Technology Kharagpur*, India
Advisor: Asst. Prof. Palash Dey

Industry Visits & Internships

Spring 2025 **Student Researcher**, *Google*
Hosts: Sreenivas Gollapudi, Kostas Kollias

Summer 2023 **Non-Degree Visiting Student**, *Toyota Technological Institute at Chicago*
Hosts: Ali Vakilian, Yury Makarychev, Julia Chuzhoy

Fall 2022 **Visiting Graduate Student**, *Simons Institute for the Theory of Computing*
Data-Driven Decision Processes program

Summer 2020 **Visiting Student Research Internship (Remotely)**, *Georgia Tech*, Atlanta, USA
Hosts: Prasad Tetali and Mohit Singh

Summer 2019 **Narendra Internship**, *Indian Institute of Science*, Bengaluru
Host: Arindam Khan

Selected Publications

Full publication list on page 3 and at mathrulestheworld.github.io.

STOC '26 Combinatorial Optimization using Comparison Oracles
with Vincent Cohen-Addad, Tommaso d'Orsi, Anupam Gupta, Guru Guruganesh, Euiwoong Lee, Renato Paes Leme, Debmalya Panigrahi, Jon Schneider, and David P. Woodruff
Optimization from Human Feedback

ICALP '25 Guessing Efficiently for Constrained Subspace Approximation
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
Constrained PCA / Dimensionality Reduction

SODA '24 Approximation Algorithms for Weighted Nash Social Welfare via Convex and Non-Convex Programs
with Adam Brown, Aditi Laddha, and Mohit Singh
Fair Resource Allocation

FOCS '23 The Price of Explainability for Clustering
with Anupam Gupta, Ola Svensson, and Rachel Yuan
Explainable Clustering

FOCS '22 Determinant Maximization via Matroid Intersection Algorithms
with Adam Brown, Aditi Laddha, Mohit Singh, and Prasad Tetali
Experimental Design / Subset Selection

SODA '22 A 3-Approximation Algorithm for Maximum Independent Set of Rectangles
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
Computational Geometry

Awards and Achievements

- Selected for the **Iliad Fellowship** for mathematical research on AI alignment, 2026
- Secured **Bronze Medal** representing India at the **57th International Mathematical Olympiad (IMO)**, 2016
- **IITKGP Foundation-USA International Internship Award**, 2020
- **Infosys Award** for excellent performance in International Olympiads, 2016
- **Best Solution Award** in IMO training camp, 2015

Complete Publication Record

Peer-Reviewed Journal Articles

2026 **A $(2+\varepsilon)$-Approximation Algorithm for Maximum Independent Set of Rectangles**
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
ACM Transactions on Algorithms, online publication, July 2, 2026.

2025 **Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs**
with Adam Brown, Aditi Laddha, and Mohit Singh
Mathematical Programming, Series A, online-first, December 23, 2025. Journal version of the SODA 2024 paper.

Peer-Reviewed Conference Papers

2026 **Combinatorial Optimization using Comparison Oracles**
with Vincent Cohen-Addad, Tommaso d'Orsi, Anupam Gupta, Guru Guruganesh, Euiwoong Lee, Renato Paes Leme, Debmalya Panigrahi, Jon Schneider, and David P. Woodruff
58th ACM Symposium on Theory of Computing (**STOC**), pp. 1995–2006.

2025 **Max-Cut with Multiple Cardinality Constraints**
with Yury Makarychev and Ali Vakilian
APPROX/RANDOM, LIPIcs 353, Article 13, pp. 13:1–13:21.

2025 **Guessing Efficiently for Constrained Subspace Approximation**
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
52nd International Colloquium on Automata, Languages, and Programming (**ICALP**), LIPIcs 334, Article 29, pp. 29:1–29:20.

2024 **Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs**
with Adam Brown, Aditi Laddha, and Mohit Singh
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), pp. 1307–1327.

2023 **The Price of Explainability for Clustering**
with Anupam Gupta, Ola Svensson, and Rachel Yuan
IEEE Symposium on Foundations of Computer Science (**FOCS**), pp. 1131–1148.

2022 **Determinant Maximization via Matroid Intersection Algorithms**
with Adam Brown, Aditi Laddha, Mohit Singh, and Prasad Tetali
IEEE Symposium on Foundations of Computer Science (**FOCS**), pp. 255–266.

2022 **A 3-Approximation Algorithm for Maximum Independent Set of Rectangles**
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke, and Andreas Wiese
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), pp. 894–905.

2020 **On Guillotine Separability of Squares and Rectangles**
with Arindam Khan
APPROX/RANDOM, LIPIcs 176, Article 47, pp. 47:1–47:22.

Additional Public Preprints

2025 **An Algorithmic Upper Bound for Permanents via a Permanental Schur Inequality**
with Aditi Laddha
Preprint, arXiv:2509.08121, 2025.

2022 **Efficient Determinant Maximization for All Matroids**
with Adam Brown, Aditi Laddha, and Mohit Singh
Preprint, arXiv:2211.10507, 2022.

Selected Talks and Presentations

2026 Combinatorial Optimization using Comparison Oracles, STOC, June 24, 2026.

2026 Geometric Methods for Combinatorial Optimization Using Comparison Oracles, NYU Geometry Seminar, March 10, 2026 (video).

2025 Combinatorial Optimization using Comparison Oracles, Yale Theory Student Seminar, October 24, 2025.

2025 Max-Cut with Multiple Cardinality Constraints, APPROX, August 12, 2025.

2025 Guessing Efficiently for Constrained Subspace Approximation, ICALP, July 10, 2025.

2023 The Price of Explainability for Clustering, FOCS, November 7, 2023; also presented at CMU Theory Lunch, October 25, 2023.

2022 Efficient Determinant Maximization for All Matroids, IISc-MSR/Bangalore Theory Seminar, Indian Institute of Science, December 23, 2022 (video).

2022 A 3-Approximation Algorithm for Maximum Independent Set of Rectangles, SODA 2022.

2020 On Guillotine Separability of Squares and Rectangles, APPROX/RANDOM, August 17, 2020 (video).